NORMALIZED BASE CHANGE OF A FLAT FINITE TYPE SCHEME OVER A NAGATA DEDEKIND RING BECOMES SMOOTH OVER THE GENERIC POINTS OF ALL FIBRES AFTER A FINITE EXTENSION

THE STACKS PROJECT AUTHORS

Editorial note. Statement span is lines 19281-19289 of sources/stacks-more-morphisms/pinned-original-source.tex; proof span is lines 19291-19355, the complete author proof (63 body lines).

Editorial note. Navigation stubs: 12 cited-but-not-included labels are supplied as non-load-bearing pointers with their pinned tags where the snapshot tag index has them.

Editorial note. extra_packages is empty on purpose: the retained preamble already loads xy with options and a second declaration clashes in the assembler wrapper.

Editorial note. Text is retained verbatim, including the author's own typos and placeholders.

Editorial note. One parenthetical detail in the final paragraph is marked details omitted (why O_{Y,y} is a local ring of the integral closure); the surrounding chain of reductions is fully written.

Allowed standard prerequisites (editorial).

- Explicit prerequisite boundary. Base change compatibility of the smooth locus (Morphisms, set-points-where-fibres-smooth), the finite-extension geometric-reducedness lemmas (Varieties, finite-extension-geometrically-reduced, generic-points-geometrically-reduced, geometrically-reduced-dense-smooth-open), the extension-of-DVRs formally smooth criterion (More on Algebra, extension-dvrs-formally-smooth), and separability-implies-smoothness (Algebra, separable-smooth); cited inputs with both main paragraphs written in the retained span.

Source reference. Original theorem 033C: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19355–19355. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 00TV: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19355–19355. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 09IK: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19260–19265. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 054X: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 6524–6530. Cited by the retained statement or

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proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 09E7: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19355–19355. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 0GLR: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19355–19355. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 09II: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19355–19355. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 01V5: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19355–19355. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 02V4: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19355–19355. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 04KT: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19355–19355. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 04KS: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19355–19355. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Source reference. Original theorem 056V: sources/stacks-more-morphisms/pinned-original-source.tex, source lines 19355–19355. Cited by the retained statement or proof but not part of the collected spans; supplied only as a navigation pointer. The argument does not depend on its content.

Theorem 0.1. *Let A be a Dedekind ring with fraction field K. Let X be a scheme flat and of finite type over A. Assume A is a Nagata ring. There exists a finite extension L/K such that the normalized base change Y is smooth over* $\mathrm{Spec}(B)$ *at all generic points of all fibres.*

Proof. During the proof we will repeatedly use that formation of the set of points where a (flat, finitely presented) morphism like $X \to \mathrm{Spec}(A)$ is smooth commutes with base change, see Morphisms, Lemma 02V4.

We first choose a finite extension L/K such that $(X_L)_{red}$ is geometrically reduced over L, see Varieties, Lemma 04KT. Since $Y \to (X_B)_{red}$ is birational we see applying Varieties, Lemma 04KS that Y_L is geometrically reduced over L as well. Hence $Y_L \to \mathrm{Spec}(L)$ is smooth on a dense open $V \subset Y_L$ by Varieties, Lemma 056V. Thus the smooth locus $U \subset Y$ of the morphism $Y \to \mathrm{Spec}(B)$ is open (by Morphisms, Definition 01V5) and is dense in the generic fibre. Replacing A by B and X by Y we reduce to the case treated in the next paragraph.

Assume X is normal and the smooth locus $U \subset X$ of $X \to \mathrm{Spec}(A)$ is dense in the generic fibre. This implies that U is dense in all but finitely many fibres, see Lemma 054X. Let $x_1, \ldots, x_r \in X \setminus U$ be the finitely many generic points of irreducible components of $X \setminus U$ which are moreover generic points of irreducible components of fibres of $X \to \mathrm{Spec}(A)$. Set $\mathcal{O}_i = \mathcal{O}_{X,x_i}$. Let A_i be the localization of A at the maximal ideal corresponding to the image of x_i in $\mathrm{Spec}(A)$. By More on Algebra, Proposition 09II there exist finite extensions K_i/K which are solutions for the extension of discrete valuation rings $A_i \to \mathcal{O}_i$. Let L/K be a finite extension dominating all of the extensions K_i/K. Then L/K is still a solution for $A_i \to \mathcal{O}_i$ by More on Algebra, Lemma 0GLR.

Consider the diagram (09IK) with the extension L/K we just produced. Note that $U_B \subset X_B$ is smooth over B, hence normal (for example use Algebra, Lemma 033C). Thus $Y \to X_B$ is an isomorphism over U_B. Let $y \in Y$ be a generic point of an irreducible component of a fibre of $Y \to \mathrm{Spec}(B)$ lying over the maximal ideal $\mathfrak{m} \subset B$. Assume that $y \notin U_B$. Then y maps to one of the points x_i. It follows that $\mathcal{O}_{Y,y}$ is a local ring of the integral closure of $\mathcal{O}_i$ in $R(X) \otimes_K L$ (details omitted). Hence because L/K is a solution for $A_i \to \mathcal{O}_i$ we see that $B_{\mathfrak{m}} \to \mathcal{O}_{Y,y}$ is formally smooth in the $\mathfrak{m}_y$-adic topology (this is the definition of being a "solution"). In other words, $\mathfrak{m}\mathcal{O}_{Y,y} = \mathfrak{m}_y$ and the residue field extension is separable, see More on Algebra, Lemma 09E7. Hence the local ring of the fibre at y is $\kappa(y)$. This implies the fibre is smooth over $\kappa(\mathfrak{m})$ at y for example by Algebra, Lemma 00TV. This finishes the proof. $\square$